STACKS, DIFFERENTIAL GRADED SHEAVES: PROPOSITION-DERIVED-COMPLETE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 6601-6618 of sources/stacks-sdga/pinned-original-source.tex; proof span is lines 6620-6722, the complete author proof (101 body lines).

Editorial note. Navigation stubs: 8 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 2 of queue_sdga.json (pb=4129, stmt 6601-6618, proof 6620-6722).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original proposition 09J1: sources/stacks-sdga/pinned-original-source.tex, source lines 6722–6722. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 0GZ5: sources/stacks-sdga/pinned-original-source.tex, source lines 6722–6722. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 0GZF: sources/stacks-sdga/pinned-original-source.tex, source lines 6358–6368. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 0GZJ: sources/stacks-sdga/pinned-original-source.tex, source lines 6557–6560. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 0GZH: sources/stacks-sdga/pinned-original-source.tex, source lines 6452–6459. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 091S: sources/stacks-sdga/pinned-original-source.tex, source lines 6722–6722. Cited by the retained statement or proof but

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not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 091W: sources/stacks-sdga/pinned-original-source.tex, source lines 6722–6722. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 0922: sources/stacks-sdga/pinned-original-source.tex, source lines 6722–6722. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Proposition 0.1. *Let R be a Noetherian ring. Let $I \subset R$ be an ideal. The following three categories are canonically equivalent:*

1. *Let $\mathcal{A}$ be the sheaf of R-algebras on $\mathbf{N}$ corresponding to the inverse system of R-algebras $A_n = R/I^n$. The category $QC(\mathcal{A})$.*
2. *Choose generators $f_1, \ldots, f_r$ of I. Let $\mathcal{B}$ be the sheaf of differential graded R-algebras on $\mathbf{N}$ corresponding to the inverse system of Koszul algebras on $f_1^n, \ldots, f_r^n$. The category $QC(\mathcal{B})$.*
3. *The full subcategory $D_{comp}(R, I) \subset D(R)$ of derived complete objects, see More on Algebra, Definition 091S and text following.*

Proof. Consider the obvious morphism $f : (Sh(\mathbf{N}), \mathcal{A}) \to (Sh(pt), R)$ of ringed topoi and let us consider the adjoint functors Lf^* and Rf_*. The first restricts to a functor

$$F : D_{comp}(R, I) \longrightarrow QC(\mathcal{A})$$

which sends an object K of $D_{comp}(R, I)$ represented by a K-flat complex $K^\bullet$ to the object $(K^\bullet \otimes_R R/I^n)$ of $QC(\mathcal{A})$. The second restricts to a functor

$$G : QC(\mathcal{A}) \longrightarrow D_{comp}(R, I)$$

which sends an object $(M_n^\bullet)$ of $QC(\mathcal{A})$ to $R\lim M_n^\bullet$. The output is derived complete for example by More on Algebra, Lemma 091W. Also, it follows from More on Algebra, Proposition 0922 that $G \circ F = \mathrm{id}$. Thus to see that F and G are quasi-inverse equivalences it suffices to see that the kernel of G is zero (see Derived Categories, Lemma 09J1). However, it does not appear easy to show this directly!

In this paragraph we will show that $QC(\mathcal{A})$ and $QC(\mathcal{B})$ are equivalent. Write $\mathcal{B} = (B_n)$ where B_n is the Koszul complex viewed as a cochain complex in degrees $-r, -r+1, \ldots, 0$. By Divided Power Algebra, Remark 0GZ5 (but with chain complexes turned into cochain complexes) we can find $1 < n_1 < n_2 < \ldots$ and maps of differential graded R-algebras $B_{n_i} \to E_i \to R/(f_1^{n_i}, \ldots, f_r^{n_i})$ and $E_i \to B_{n_{i-1}}$ such that

$$\begin{array}{ccccccc}
B_{n_1} & \longleftarrow & B_{n_2} & \longleftarrow & B_{n_3} & \longleftarrow & \ldots \\
\downarrow & & \downarrow & & \downarrow & & \\
E_1 & \longleftarrow & E_2 & \longleftarrow & E_3 & \longleftarrow & \ldots \\
\downarrow & & \downarrow & & \downarrow & & \\
B_1 & \longleftarrow & B_{n_1} & \longleftarrow & B_{n_2} & \longleftarrow & \ldots
\end{array}$$

is a commutative diagram of differential graded R-algebras and such that $E_i \to R/(f_1^{n_i}, \ldots, f_r^{n_i})$ is a quasi-isomorphism. We conclude

1. there is an equivalence between $QC(\mathcal{B})$ and $QC((E_i))$,
2. there is an equivalence between $QC((E_i))$ and $QC((R/(f_1^{n_i}, \ldots, f_r^{n_i})))$,

(3) there is an equivalence between $QC((R/(f_1^{n_i}, \ldots, f_r^{n_i})))$ and $QC(\mathcal{A})$.

Namely, for (1) we can apply Lemma 0GZH to the diagram above which shows that (E_i) and (B_n) are pro-isomorphic. For (2) we can apply Lemma 0GZF to the inverse system of quasi-isomorphisms $E_i \to R/(f_1^{n_i}, \ldots, f_r^{n_i})$. For (3) we can apply Lemma 0GZH and the elementary fact that the inverse systems (R/I^n) and $(R/(f_1^{n_i}, \ldots, f_r^{n_i})$ are pro-isomorphic.

Exactly as in the first paragraph of the proof we can define adjoint functors[1]

$$F' : D_{comp}(R, I) \longrightarrow QC(\mathcal{B}) \quad \text{and} \quad G' : QC(\mathcal{B}) \longrightarrow D_{comp}(R, I).$$

The first sends an object K of $D_{comp}(R, I)$ represented by a K-flat complex $K^\bullet$ to the object $(K^\bullet \otimes_R B_n)$ of $QC(\mathcal{B})$. The second sends an object (M_n) of $QC(\mathcal{B})$ to $R\lim M_n$. Arguing as above it suffices to show that the kernel of G' is zero. So let $\mathcal{M} = (M_n)$ be a good sheaf of differential graded modules over $\mathcal{B}$ which represents an object of $QC(\mathcal{B})$ in the kernel of G'. Then

$$0 = R\lim M_n \Rightarrow 0 = (R\lim M_n) \otimes^{\mathbf{L}}_R B_t = R\lim(M_n \otimes^{\mathbf{L}}_R B_t)$$

By Lemma 0GZJ we have $R\lim(M_n \otimes^{\mathbf{L}}_R B_t) = R\lim(M_n \otimes^{\mathbf{L}}_{B_n} B_t)$. Since (M_n) is an object of $QC(\mathcal{B})$ we see that the inverse system $M_n \otimes^{\mathbf{L}}_{B_n} B_t$ is eventually constant with value M_t. Hence $M_t = 0$ as desired. □

[1]It can be shown that these functors are, via the equivalences above, compatible with F and G defined before.